INCH-POUND

MIL-S-19500/399A
19 June 1990
SUPERSEDING
MIL-S-19500/399
15 August 1968

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPE 2N3960, JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, switching transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
mW	V dc	V dc	V dc	°C
400	20	12	4.5	-65 to +200

1/ Derate linearly 2.30 mW/°C for T_A = 25°C.

1.4 Primary electrical characteristics.

	h_{FE}		$V_{CE(sat)}$		C_{obo}	$\|h_{fe}\|$	V_{BE}	
Limits	V_{CE} = 1.0 V dc I_C = 1.0 mA dc	V_{CE} = 1.0 V dc I_C = 10 mA dc	I_C = 1.0 mA dc I_B = 0.1 mA dc	I_C = 30 mA dc I_B = 3.0 mA dc	V_{CB} = 4.0 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 4.0 V dc I_C = 5.0 mA dc f = 100 MHz	I_C = 1.0 mA dc V_{CE} = 1.0 V dc	I_C = 30 mA dc V_{CE} = 1.0 V dc
			V dc	V dc	pF		V dc	V dc
Min	40	60	---	---	---	13	---	---
Max	---	300	0.2	0.3	2.5	---	0.8	1.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBRO)), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold-plated, tin-plated, or solder-dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

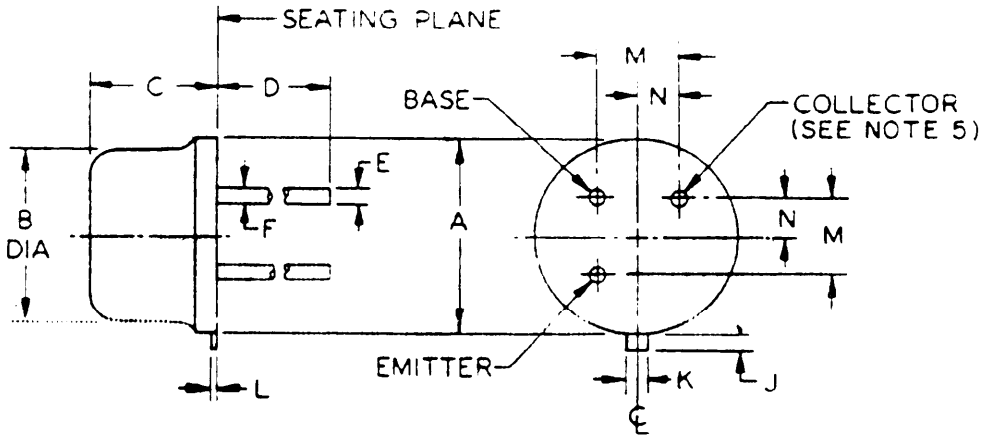
3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the country of origin, and manufacturer's identification may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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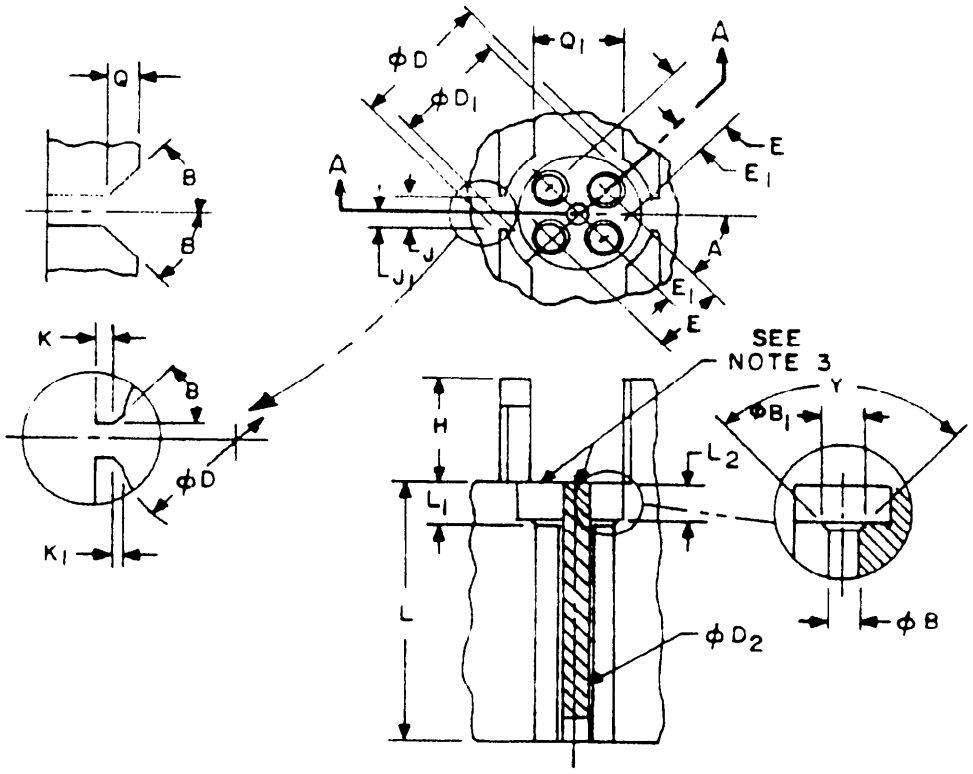


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.209	.230	5.31	5.84	
B	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	.750	12.70	19.05	7
E	.016	.021	0.41	0.53	2,7
F	.016	.019	0.41	0.48	3,7
J	.028	.048	0.71	1.22	6
K	.036	.046	0.91	1.17	
L	---	.020	---	0.51	
M	.0707 Nom		1.796 Nom		4
N	.0354 Nom		0.899 Nom		4

NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone between .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in gauging plane .054 + .001, -.000 (1.37 + 0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred measurement method.
5. The collector shall be electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All three leads.

FIGURE 1. Physical dimensions.

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SECTION A-A

FIGURE 2. Gauge for lead and tab location.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
øB	.0325	.0335	0.826	0.851	4
$øB_1$	.043 Nom		1.09 Nom		4
øD	.2310	.2315	5.87	5.88	
$øD_1$	.159	.161	4.04	4.09	
$øD_2$	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.527	2.553	
E_1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.206	
J_1	.0235	.0245	0.597	0.622	

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
K	.009	.011	0.23	0.28	
K_1	.005 Nom		0.13 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90°	45.10°	---	---	
B	45° Nom				
Y	90° Nom				

NOTES:
1. Metric equivalents are given for general information only.
2. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) ±.010 (0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gauge for lead and tab location - Continued.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-STD-19500)	Measurement
	JANTX and JANTXV levels
9	Not applicable
10	V_{CB} = 12 V
11	I_{CEX2} and h_{FE2}
12	See 4.3.1
13	Subgroup 2 of table I herein, ΔI_{CEX2} = 100 percent of initial value or 20 nA dc, whichever is greater; Δh_{FE2} = 20 percent of initial value

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = 25°C ±3°C; V_{CB} = 16 V dc; P_T = 400 mW.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an RF signal voltage of 1.0 volt (rms) across the collector-base terminals, and measuring the ac voltage drop (V_{EB}) with a high impedance rf voltmeter across the emitter-base terminals. With f = 31.8 MHz used for the 1.0 volt signal, the following computation applies: $r_b'C_c$ (picoseconds) = 5 x V_{EB} (millivolts).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3001	Bias condition D, I_C = 10 µA dc	$V_{(BR)CBO}$	20		V dc
Breakdown voltage, emitter to base	3026	Bias condition D, I_C = 10 µA dc	$V_{(BR)EBO}$	4.5		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 10 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	12		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 10 V dc, V_{BE} = 0.4 V dc	I_{CEX1}		1.0	µA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 10 V dc, V_{EB} = 2.0 V dc	I_{CEX2}		5.0	nA dc
Forward current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 1.0 mA dc	h_{FE1}	40		
Forward current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE2}	60	300	
Forward current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 30 mA dc pulsed (see 4.5.1)	h_{FE3}	30		
Collector to emitter saturation voltage	3071	I_C = 1.0 mA dc, I_B = 0.1 mA dc	$V_{CE(sat)1}$		0.2	V dc
Collector to emitter saturation voltage	3071	I_C = 30 mA dc, I_B = 3.0 mA dc pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.3	V dc
Base emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 1.0 mA dc	V_{BE1}		0.8	V dc
Base emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 30 mA dc	V_{BE2}		1.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 10 V dc, V_{EB} = 2 V dc	I_{CEX3}		5.0	µA dc
Low-temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 1.0 V dc, I_C = 10 mA dc pulsed, (see 4.5.1)	h_{FE4}	30		
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc, I_C = 5.0 mA dc, f = 100 MHz	$\lvert h_{fe1} \rvert$	13	29	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc, I_C = 10 mA dc, f = 100 MHz	$\lvert h_{fe2} \rvert$	14	32	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc, I_C = 30 mA dc, f = 100 MHz	$\lvert h_{fe3} \rvert$	12	29	
Open circuit output capacitance	3236	V_{CB} = 4 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		2.5	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		2.5	pF
Collector-base time constant	---	V_{CE} = 4 V dc, I_C = 10 mA dc, f = 31.8 MHz, See 4.5.2 and figure 3.	$r_b'C_c$	0	50	ps
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group B inspection for JANTX and JANTXV devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1 2/		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	
Hermetic seal	1071	
a. Fine leak b. Gross leak		
Electrical measurements		See table IV, steps 1 and 3.
Subgroup 3		
Steady-state operation life	1026	V_{CB} = 10 V dc, P_T = 400 mW ±20 mW T_A = 25°C ±5°C.
Electrical measurements		See table IV, steps 2 and 3.
Subgroup 4		
Decap internal visual (design verification)	2075	
Bond strength	2037	Test condition A
Subgroup 5		
Not applicable		
Subgroup 6		
High-temperature life (non-operating)	1032	T_{STG} = 200°C
Electrical measurements		See table IV, steps 2 and 4.

1/ For sampling plan, see MIL-S-19500.

2/ A separate sample may be pulled for each test.

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TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1.
Subgroup 2		
Thermal shock (glass strain)	1056	Test condition A
Therminal strength	2036	Test condition E
Hermetic seal	1071	
a. Fine leak b. Gross leak		
Moisture resistance	1021	
Electrical measurements		See table IV, steps 1 and 3.
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	
Constant acceleration	2006	
Electrical measurements		See table IV, steps 1 and 3.
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Steady-state-operation life	1026	T_A = 25°C ±5°C, V_{CB} = 10 V dc, P_T = 400 mW ±20 mW
Electrical measurements		See table IV, steps 2 and 3.
Subgroup 7		
Not applicable		

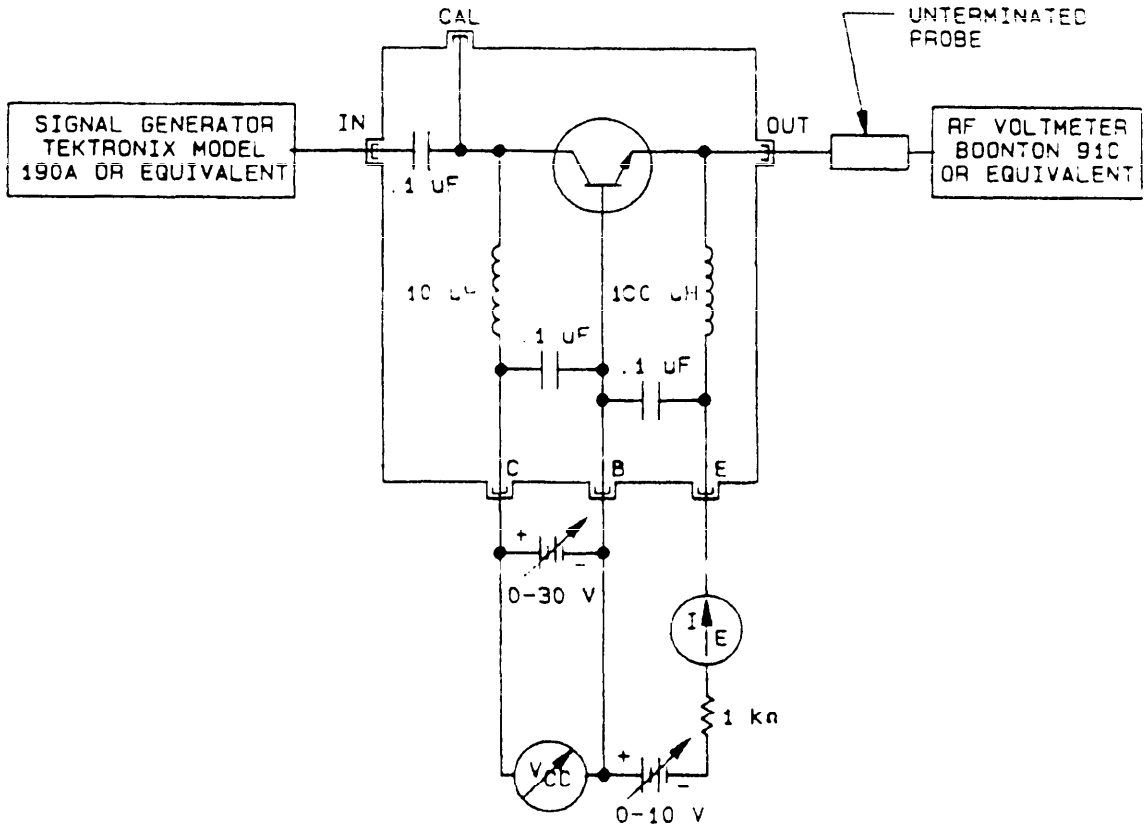
1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1	Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 10 V dc, V_{EB} = 2 V dc	I_{CEX1}		5.0	nA dc
2	Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 10 V dc, V_{EB} = 2 V dc	I_{CEX1}		10	nA dc
3	Forward current transfer ratio	3076	V_{CE} = 1.0 V dc, I_C = 10 mA dc pulsed (see 4.5 1)	h_{FE2}	60	300	
4	Forward current transfer ratio	3076	V_{CE} = 1.0 V dc, I_C = 10 mA dc pulsed (see 4.5.1)	Δh_{FE2}	20 percent change from initial value		

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Procedure.

1. Set signal generator to 31.8 MHz and connect to "IN" connector on test jig.
2. Connect low voltage dc power supplies as shown. A 1 kilohm resistor should be placed in series with the emitter power supply to prevent damage to transistors being tested.
3. Place transistor to be tested in socket. Set collector supply for V_{CE} = 4.0 V dc, and emitter supply for I_C as specified.
4. Connect RF voltmeter with unterminated probe adapter to "CAL" connector on test jig. Adjust signal generator until RF voltage is 1.00 V. (Note: Decade switching of voltmeter should be accurate from 1 mV to 3 V. If not, input voltage may be set using voltage dividers, utilizing lower scale of the RF voltmeter. If this is done, the voltage dividers should be left in place when the voltmeter is removed, as they constitute a load on the input of the circuit.)
5. Remove RF voltmeter from "CAL" connector and connect to "OUT" connector. Meter will now read $r_b'C_C$ as follows:

Meter range full scale	$r_b'C_C$ range (picoseconds)
.003 V	5 to 15 ps
.01 V	15 to 50 ps
.03 V	50 to 150 ps

FIGURE 3. Collector-base time constant test circuit (an equivalent circuit may be used).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents may specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish may be specified (see 3.3.1).

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturers' and users' Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturers' CAGE code	Manufacturers' and users' PIN
2N3960JTX	04713	SRF5218H
2N3960JTXV	04713	SRFE1032H SRF1353H SRF2273H, ML0203-0093 SRF3698H, 8A39-001CHD SRF5277H, 8A39-01CHA SRF671H, 48-T05996F201 SRF7052H, 169869-101

6.3.1 JAN replacement. For JAN replacement see MIL-S-19500, section 1.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 85
DLA - ES

User activities:
Army - AV, SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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